User Churn Project | Regression Modeling Results

Prepared for: Waze Leadership Team

OVERVIEW

This project analyzed user retention and churn patterns within the Waze dataset to identify key behavioral drivers of user loss. Using exploratory data analysis, feature engineering, and logistic regression modeling, the goal was to evaluate how driving patterns and app engagement impact long-term retention.

PROJECT STATUS

Milestone 5 - Regression Modeling

🎯 **Target Goal:** Build an explanatory model to identify why Waze users churn and predict which users are at risk of leaving the app.

🎯 **Methods:**

Conducted data cleaning, handled missing values, capped extreme outliers, engineered new engagement features, and built a baseline binomial logistic regression model.

🎯 **Impact:** Delivered initial actionable insights on key retention drivers (e.g., active days vs. driven distance). However, because recall for churned users remains low, the model serves as an explanatory baseline rather than an automated prediction system.

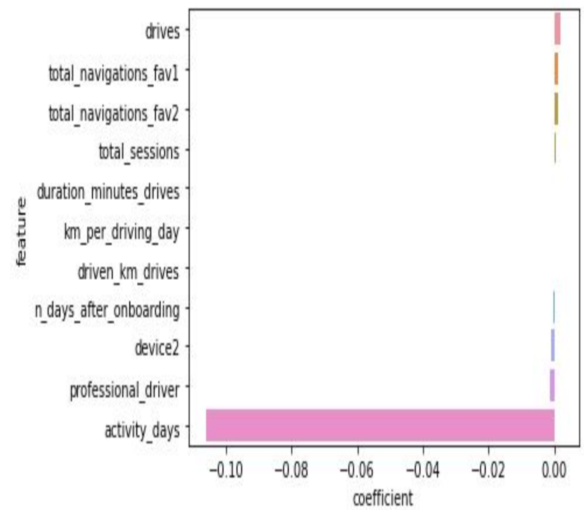
NEXT STEPS

➔ **Upgrade Modeling Techniques:** Train non-linear tree-based models (such as Random Forest or XGBoost) to improve predictive accuracy and better capture complex user behavior patterns.

➔ **Address Class Imbalance:** Apply resampling techniques like SMOTE or adjust class weights to improve model recall for churned users.

➔ **Shift Business Focus:** Pivot retention campaigns toward building regular daily app habits rather than targeting long-distance users.

KEY INSIGHTS



- **Frequency Drives Retention:** The number of active days (activity_days) and driving days (driving_days) per month are the strongest indicators of user retention. Users with consistent daily activity are significantly less likely to churn.
- **Distance Does Not Impact Churn:** Total distance driven (driven_km_drives) and total driving duration show almost no correlation with churn, meaning long-distance drivers churn at similar rates to short-distance drivers.
- **Feature Redundancy & Non-Linearity:** High correlation between features (such as drives vs. sessions) and non-linear user behavior limit the effectiveness of standard logistic regression for prediction.